

ON OPTIMAL SMOOTHING AND DERIVATIVE ESTIMATION FROM NOISY DISPLACEMENT DATA IN BIOMECHANICS *

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Estimating derivatives from noisy displacement data is a notoriously *ill-posed* problem in signal processing and biomechanics. Following the work of Wood and Jennings (1978) and Hatze (1979, 1981), the present paper describes the use of optimally regularized, natural quintic splines for estimating smoothed positions, velocities, and accelerations from equidistantly sampled, noisy position measurements. It appears that the nature of the boundary conditions of the data is of some importance, since various algorithms used hitherto result in artefacts throughout the data if the true derivatives at the record ends differ significantly from zero. Natural quintic splines do not suffer from this disadvantage below the third derivative.

The ill-posed character of movement analysis has some interesting implications for movement synthesis and optimization, similar to the indeterminacy of muscular co-contraction from merely external, biomechanical measurements.

Introduction

In studies of human movement (see Hatze (1984) for a review), evaluating the kinetics of movement and the load distribution within the musculoskeletal components generally requires estimates of the inertial and gravitational force contributions. This necessitates both the assess-

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ment of the mass distributions in the body segments under consideration, and estimation of the translational and rotational displacements, velocities, and accelerations of these segments. In the analysis of slow, pathological gait, inertial (and gravitational) force contributions are usually neglected, and the net joint forces and moments are estimated 'from the ground up', based on force-plate recordings and displacement data only. As exemplified by the data in Bresler and Frankel (1950: fig. 9), Wells (1981), and Boccardi et al. (1981), this approach is acceptable at the ankle, less acceptable at the knee, and inappropriate at the hip, for the case of ordinary, level locomotion. In jerky, pathological gait, and in fast, sportive movement, neglect of inertial effects can be expected to be unjustifiable.

Most equipment for kinematic movement analysis (see the recent first issue of the I.S.B.S. *Sports Biomechanics Newsletter*, or the review in Woltring (1984)) measures position or direction data on a limited set of artificial landmarks affixed to the body segments. The problem now is to estimate the first and second derivatives of these data (or of their 3-D transformations), given the usually noisy nature of these measurements. If the power spectra of the signal and noise were known, the classical Wiener filter (Wiener 1949) would be appropriate for least-squares optimal separation of the signal from the noise. Unfortunately, such information is usually not available in kinesiology and biomechanics, while it appears that estimated derivatives are very sensitive to typical measurement errors and to selected filter parameters.

In the terminology of statistics and mathematical physics, derivative estimation from noisy measurements is a (moderately) *ill-posed problem*, together with a variety of other mathematical problems in such diverse fields as computerized tomography, electrocardiology, meteorology, and seismology. The term *ill-posed* (Hadamard 1902) refers to the fact that the quantities being estimated are highly sensitive to certain types of errors in the measurements. Ill-posedness is intrinsic to the problem under study, as opposed to *ill-conditioning* of the numerical equations ensuing from quantifying a problem; the latter is a rounding error effect caused by finite precision in computer arithmetic. A simple illustration of ill-posedness in differentiation follows from assuming that the measurements contain an additive sinusoidal term $A \cdot \sin(\omega t)$, with very small amplitude A , and very large frequency ω . In the measured data, the influence of this term is negligible, but the amplitude ωA of the differentiated error term $\omega A \cdot \cos(\omega t)$ is very large

for sufficiently large ω (and even more so for the higher derivatives). The same occurs if the data are contaminated with wide-band measurement noise, which can be modelled in terms of a sum of many low-level, high-frequency sinusoids.

It appears that it is necessary to include additional constraints, in order to arrive at meaningful derivative estimates. A suitable criterion is to assume that the movement is sufficiently *smooth*, i.e., it does not contain inordinately high frequency components, since the opposite would entail too high inertial forces. This condition can be translated into the requirement that the estimated derivatives of the data stay within reasonable bounds. In effect, the goal should be to search for smooth estimates which at the same time are sufficiently consistent with the measurements. If the ensuing 'optimal' estimates (using any of the algorithms discussed below) appear too smooth, this may provide sufficient reason to improve the spatio-temporal resolution of the measuring equipment; see Lanshammar (1982a, 1982b) and Woltring (1984). In other words, the sampling frequency must be sufficiently high to ensure that the signal and noise have negligible components outside the Nyquist band, as prescribed by the Shannon-Whittaker sampling theorem. In the opposite case, high-frequency components will be folded back into the Nyquist band, and result in signal distortion or increased noise level.

Many authors in biomechanics (e.g., Winter et al. 1974; Lesh et al. 1979) have attempted to use existing, standard digital signal processing methods for obtaining suitable derivatives; see Wood (1982) for a review. Usually, a number of filter parameters must be set by the user through trial-and-error in order to strike a reasonable balance between signal distortion and noise transmission. For frequency-domain methods, this implies the selection of the position and shape of the transition band between low-pass signal transmission and high-frequency cut-off; for spline functions, one should select the type of the spline and a suitable 'fiddle factor' S , as discussed below.

In recent years, much effort has been spent on automatic determination of optimal filter parameters *from the measurements*. Wood and Jennings (1978) were among the first in the published biomechanical literature to adopt this approach. They used cubic and quintic (polynomial) spline functions, and Soudan and Dierckx (1979) referred to the work of Wahba and Wold (1975a) on spline 'Cross-Validation' which is one particular form of *optimal regularization* as discussed

below. Hatze (1979, 1981) investigated a related approach where the data are processed via a Fast Fourier Transform procedure (cf. Gentleman and Sande 1966), based on the studies of Cullum (1971) and Anderssen and Bloomfield (1974a, 1974b). He assumed that the data are weakly stationary after linear detrending. Wahba and Wold (1975b) provided a comparison between these methods (cf. also Cullum 1979). More recently, Ahlén (1984) used autoregressive/moving-average and state-space models for optimally estimating higher derivatives from noisy data sequences. In the following paragraphs, the use of Utreras' fast approximation of the Generalized Cross-Validation Criterion (Craven and Wahba 1979; Golub et al. 1979; Utreras 1979, 1980, 1981) for optimally regularizing cubic and quintic, natural splines is discussed, and compared with Hatze's [1] FFT-approach. Both algorithms assume that the data have been sampled equidistantly, and that the noise is additive, uncorrelated, zero-mean, and stationary.

Optimal regularization

The term *regularization* refers to an approach where high-frequency oscillations in an ill-posed problem are damped out by including a penalty on some transformation of the smoothed data, e.g., their m -th derivative (or a combination of different derivatives). Two forms are practised; the first one requires that the selected norm is minimized, provided that the (weighted) rms error between the raw data $\{g(t_i)\}$ and the smoothed data $\{s(t_i)\}$ does not surpass a given tolerance level $S \geq 0$,

$$\min_{s(t)} \int_{t_1}^{t_n} [s^{(m)}(t)]^2 dt; \quad \sum_{i=1}^n |s(t_i) - g(t_i)|^2 \leq S. \quad (1)$$

The second form relies on a weight parameter μ , and requires minimization of the sum of the above two terms,

$$\min_{s(t)} \left[\mu \cdot \int_{t_1}^{t_n} [s^{(m)}(t)]^2 dt + \sum_{i=1}^n |s(t_i) - g(t_i)|^2 \right], \quad (2)$$

for suitably selected $\mu > 0$. Note that the measurements are assumed to be available as discrete samples, at time instants t_i , $i = 1, \dots, n$.

[1] Available from BIOMLIB, P.O.B. 680765, D-8000 Munich 68, FRG.

It is known that (1) and (2) both result in *smoothing polynomial splines* of order $2m$, for given values of $S > 0$ or $\mu > 0$, where S and μ are uniquely related; see, for example, Reinsch (1971). The values $S = 0$ or $\mu = 0$ result in *interpolating splines*. The splines consist of local polynomials of degree $\leq 2m - 1$; these polynomials are piecewise continuous at the *knots* $\{t_i\}$ up to and including the $(2m - 2)$ nd derivative. For a unique estimate $s(t)$, certain boundary conditions can be imposed on $s^{(k)}(t_1)$ and $s^{(k)}(t_n)$, with $k = m, m + 1, \dots, 2m - 2$. If these values are defined as zero, the so-called *natural splines* of degree $2m - 1$ are generated. The case $m = 2$ corresponds to cubic splines, $m = 3$ to quintic splines, and $m = 4$ to heptic splines.

Formulation (2) is usually denoted as the *regularized* version of the spline estimation problem, and the question is how to select a suitable value for the *regularizing parameter* μ . Too large a value for μ (or S) will result in gross oversmoothing of the measurements, whereas too small a value will entail too much noise transmission, with deleterious effects on the estimated derivatives $s^{(k)}(t)$, $k = 1, \dots, m$. Since the splines are constrained at the m -th derivative, it is advisable to attach significance to spline derivatives of lower orders only; cf. Ragozin (1983).

Wahba and Wold (1975a, 1975b) introduced a statistical method for selecting an optimal value for μ , which is called the method of (ordinary) Cross-Validation. Basically, they fitted a spline function $f_j(t)$ to all the data, except for the measurement at time t_j . Subsequently, they assessed the fitting error in this point t_j , and defined the optimal value for μ as that value, for which the rms sum of these fitting errors attains a minimum, for $j = 1, \dots, n$.

In subsequent studies, Wahba and her collaborators used the *Generalized Cross-Validation Criterion* (GCVC) described in Golub et al. (1979). Craven and Wahba (1979) applied the GCVC to spline functions, and reported some promising results. However, a disadvantage of their GCVC is the large amount of computational effort. A fast approximation to the GCVC, for a sufficient number (e.g., ≥ 40) of equidistantly sampled data and stationary, uncorrelated, zero-mean noise has been described by Utreras (1979, 1980, 1981). For natural cubic splines, the algorithms of Craven and Wahba and Utreras are

[2] IMSL, International Mathematical and Statistical Libraries, Houston (Texas), as of Edition 8 (June 1980). This library is available in many scientific computer centres.

available as FORTRAN subroutine ICSSCV of IMSL [2]. More recently, Hutchinson and de Hoog (1985) have described a fast, general algorithm for the GCVC, and a cubic spline FORTRAN implementation CUBGCV is available from the authors upon request. A general package for polynomial splines of arbitrary order is in preparation (Hutchinson pers. comm.).

Boundary effects

Natural cubic splines impose that the second derivative vanish at the record boundaries, similar to the optimal regularization method of Anderssen and Bloomfield (1974a, 1974b) used by Hatze (1979, 1981). As indicated by Hatze (1979: 11), non-zero accelerations at the record boundaries '...can lead to erroneous derivative estimates at and near the end points of the data sequence. It is therefore advisable to treat the computed end point values with caution'. The data of Pezzack et al. (1977) which are often used for testing differentiation software in biomechanics approximately meet this boundary condition, and Hatze showed excellent results for the estimated, second derivatives. In Hatze (1981), the same data were processed with added noise, as the Pezzack et al. (1977) data are much more accurate than is habitually the case in biomechanics (cf. Soudan and Dierckx 1979; Lanshammar 1982a, 1982b); also here, the estimated acceleration data were in close agreement with the given, true accelerations, albeit with reduced bandwidth.

A different situation occurs for strongly non-zero boundary accelerations. Vaughan (1982) described the simple case of vertical position data on a freely falling golf-ball, and he demonstrated that the estimated second derivatives for a number of algorithms were far from constant. The best results were obtained when he used natural quintic splines, where the factor S of (1) was set by trial-and-error. It is to be expected that natural quintic splines are suitable for this movement paradigm, since these splines impose that the *third* derivative (jerk) vanish at the record boundaries. Under a constant acceleration regime, this is clearly the case, and boundary effects must be expected if the third derivative at the record ends is strongly non-zero. For lower derivatives, the influence of such artefacts will be small (cf. Ragozin 1983).

Experimental and simulated results

Vaughan (1982) had to find the 'fiddle factor' S in (1) by trial-and-error, and it seemed interesting to verify the optimally regularizing algorithms of Anderssen and Bloomfield (1974a, 1974b) and of Utreras (1979, 1980, 1981) on his data. In Utreras (1980), a FORTRAN package for optimally regularized estimation of cubic and quintic, natural splines is described. Both algorithms assume equidistantly sampled data, and additive, zero-mean, uncorrelated, stationary noise, with unknown variance.

The original FORTRAN implementation in REAL*8 of Utreras' quintic spline algorithm was found to suffer from numerical instabilities, particularly when executing under IBM's VM/CMS. This appeared to be caused by explicit submatrix inversion in the actual spline calculation routine (cf. de Boor 1978: 248–249). Use of an orthonormal triangularization procedure (Gentleman 1973) without explicit matrix inversion rendered the package more robust. Even better results can be obtained if the original package is operated in REAL*16 under VM/CMS, at the expense of a costly increase in CPU-time since the IBM machine emulates all REAL*16 operations in software (De Lange pers. comm.).

In fig. 1, the second derivatives estimated from Vaughan's data are

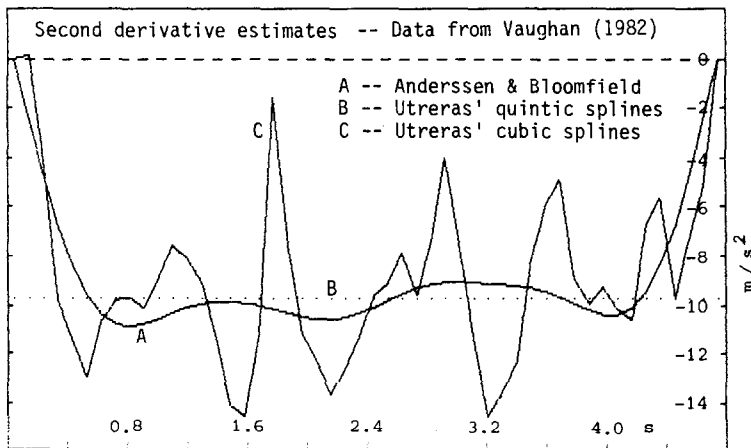


Fig. 1. Estimated second derivatives for the data of Vaughan (1982); the cubic spline of Utreras (1980) was assessed by means of IMSL routine ICSSCV, with parameter IJOB = 2.

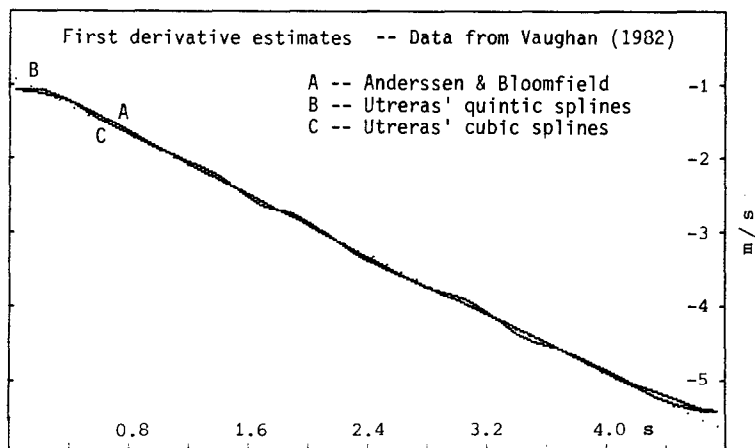


Fig. 2. Estimated first derivatives for the data of Vaughan (1982); cf. fig. 1.

shown. Clearly, the cubic spline data are worst, the quintic spline data are virtually ideal, and the regularizing FFT-approach takes an intermediate position. The quintic splines resulted in acceleration estimates which were constant up to 6 decimals, and in agreement with the value of -9.778 m/s^2 as obtained through a parametric estimation procedure. This value is slightly different from the true, local acceleration of -9.802 m/s^2 as quoted by Vaughan (1982), which can be partially

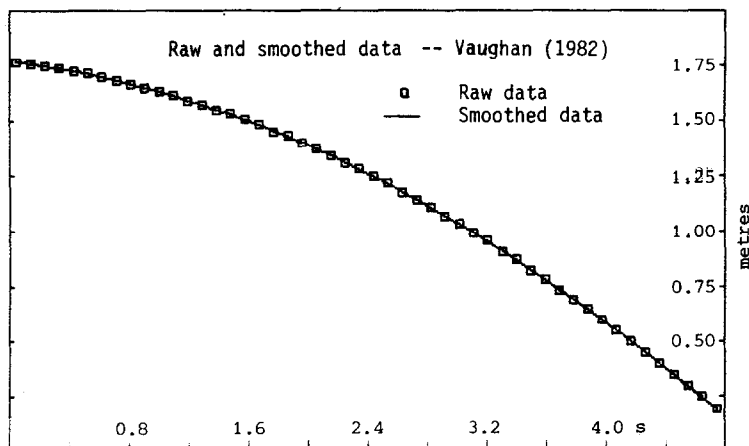


Fig. 3. Raw and smoothed data for the constant acceleration data of Vaughan (1982); cf. fig. 1.

attributed to the rounding accuracy in Vaughan's value of 9.85 ms for the sampling interval.

The lower derivatives show a much better fit for all three procedures, as becomes apparent from fig. 2 for the estimated velocities, and from fig. 3 for the raw and smoothed displacements. The FFT-algorithm was operated in the 'consistent' mode where exact derivatives are obtained from the smoothed data (cf. Hatze 1979, 1981), in order to be compatible with the spline algorithm.

In fig. 4, the estimated second derivatives of the Pezzack et al. (1977) data are shown. Here, only the FFT-method and Utreras' quintic spline procedure were used (the cubic spline yields piece-wise linear acceleration estimates). The correspondence is excellent, except at the record boundaries.

In fig. 5, the same is done for the more noisy version of the Pezzack et al. (1977) data published by Lanshammar (1982a). Both methods result in stronger smoothing, with the FFT method smoothing somewhat more than the quintic spline method.

In view of the promising results of the quintic spline approach, the influence of boundary constraint violations in the third derivative was investigated by means of synthetic data. In fig. 6, the smoothed spline data and its first three derivatives are plotted for the cubic function $g(t) = 5 \cdot t^2(1 - t/6)$ m over the range $0 \leq t \leq 2$ (s), and for additive, pseudo-gaussian noise with standard deviations 10^{-3} m and 10^{-4} m, respectively (the same noise was used in these two realizations). The data were sampled equidistantly over 50 samples, and the smoothed data were interpolated by a factor 10. The true data have constant third derivatives of -5 m/s^3 throughout the record, and are thus incompatible with the boundary conditions of natural quintic splines. By contrast, the data in fig. 7 are based on the function $g(t) = 5 \cdot t^2(1 - t^2/12)$ m for $0 \leq t \leq 1$, which is continued into the range $1 < t \leq 2$ after change of sign in the fourth derivative at $t = 1$ s, under the same noise and sampling conditions. Now, the true data have zero third derivatives and non-zero fourth derivatives at the record ends. Clearly, the quality of the highest derivatives in fig. 7 is much better than of those in fig. 6, while the lower derivatives and the smoothed data have approximately the same quality and shape.

The quintic spline routines are implemented in double precision, and require 10 times the data record length for intermediate calculations. The Anderssen and Bloomfield algorithm, on the other hand, was

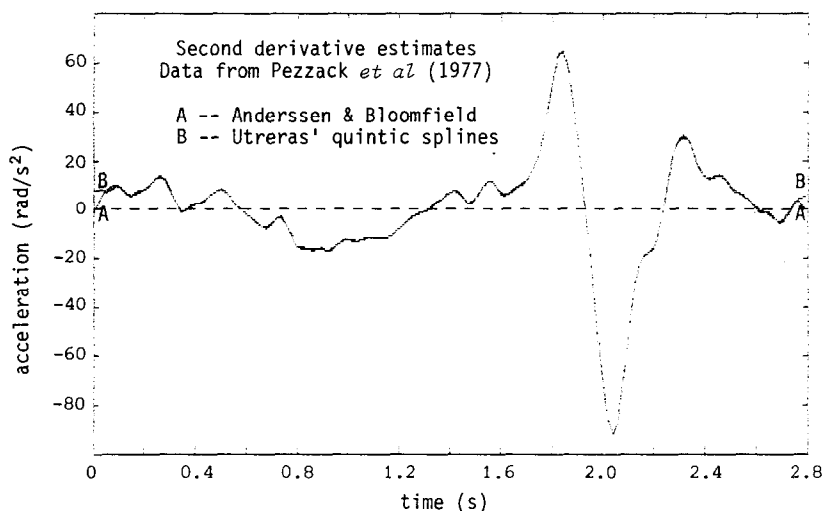


Fig. 4. Estimated second derivatives for the data of Pezzack *et al.* (1977).

implemented in single precision (Woltring and Huijses 1985), and requires ≥ 16 times the record length as working space. As regards computation time, it was found that the Anderssen and Bloomfield

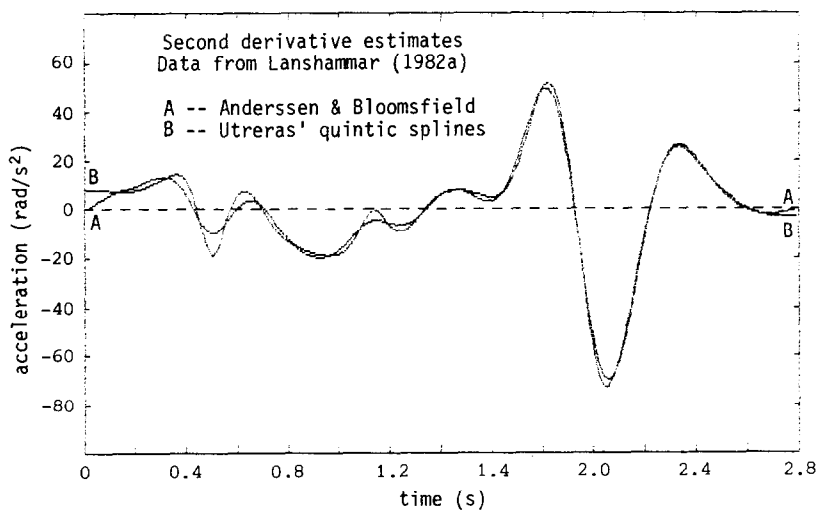


Fig. 5. Estimated second derivatives for the data of Pezzack *et al.* (1977), with additional noise as published by Lanshammar (1982a).

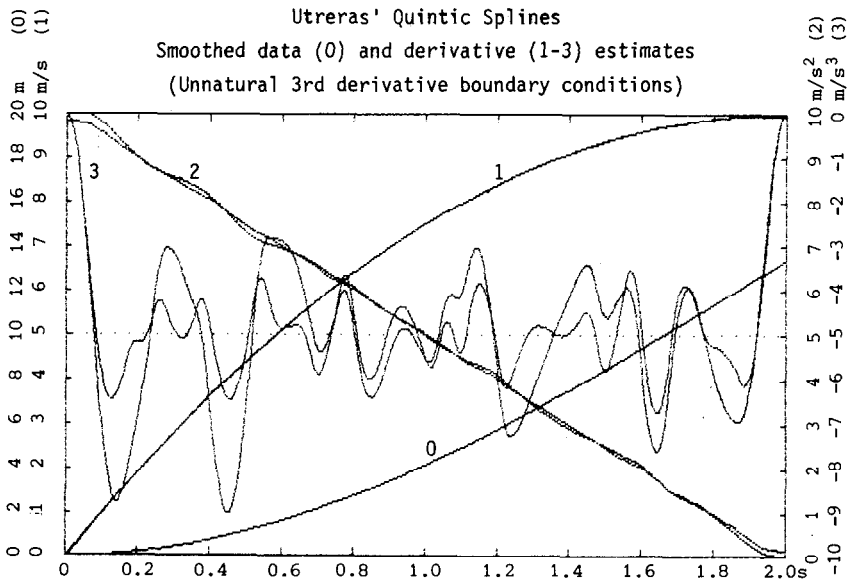


Fig. 6. Smoothed quintic spline data and estimated derivatives for synthetic, noisy data with non-zero third derivatives at the record boundaries, as explained in the text.

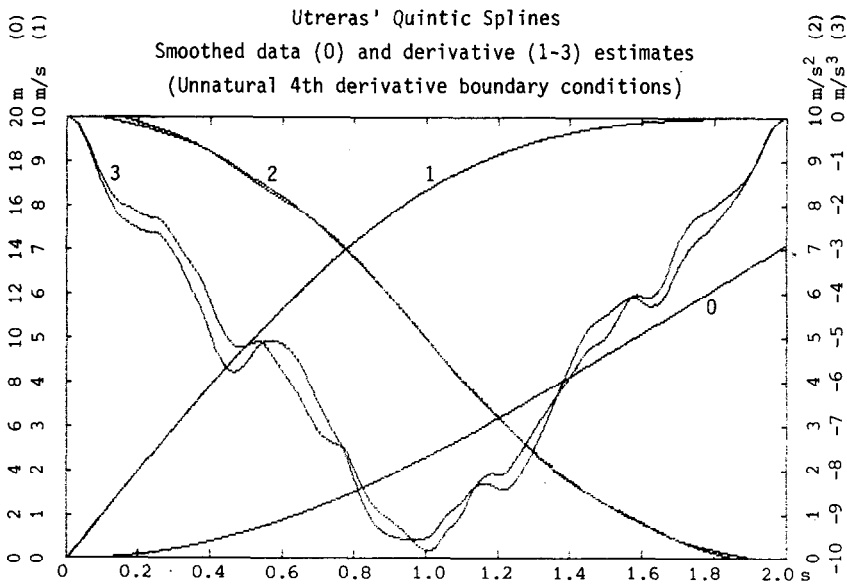


Fig. 7. Smoothed quintic spline data and estimated derivatives for synthetic, noisy data with zero third derivatives and non-zero fourth derivatives at the record boundaries, as explained in the text.

algorithm was about three times faster than Utreras' original quintic spline package. On a VAX11/780 without Floating-Point Accelerator, and with no other user processes active, the 142 data points of Pezzack et al. (1977) required 3 seconds for the Anderssen and Bloomfield algorithm, and about 10 seconds for the original quintic spline package. The package as modified for increased numerical stability required about 25 seconds for these data. Under IBM's VM/CMS, the modified package required less than one second for Vaughan's 50 data points. Computation time is approximately linear in the record length for both algorithms. One should note, though, that Utreras' algorithm is only valid for sufficiently large record length, e.g., $n \geq 40$.

Discussion

It appears that sufficiently strong boundary effects can have considerable influence throughout the data, and not merely at the record ends. A theoretical and experimental study on such effects in (cubic) spline calculus has been provided by Rice and Rosenblatt (1983). The data in figs. 6 and 7 extend their results to quintic, natural splines, since the estimated third derivatives are much smoother under more natural boundary conditions. In the case of the Anderssen and Bloomfield algorithm, boundary effects are merely due to the way that the raw data are forced into the format of the Fast Fourier Transform: the data are linearly detrended, and extended via an odd sine extrapolation in order to render them periodic as required under the FFT regime. By consequence, Vaughan's constant acceleration paradigm results in an acceleration stepfunction at the record boundaries, and any form of FFT-smoothing on such data, whether through optimal regularization or otherwise, is likely to cause oscillations throughout the record. A more appropriate approach would be to use any of the recently published methods on band-limited signal extrapolation (e.g., Sanz and Huang 1983; Byrne and Fitzgerald 1984).

Anderssen and Bloomfield (1974a, b) and Hatze (1979, 1981) derived their algorithm under the assumption that the linearly detrended, raw data are generated by a weakly stationary, stochastic process. Hatze claimed that this assumption '... is valid for data sequences observed in motion analysis and many other biomechanical investigations' (Hatz 1981: 14). However, practical work in biomechanics is often concerned

with short data records (e.g., joint excursions during a single step on a force plate) which cannot be viewed as weakly stationary, even after linear detrending. It is for such types of data that optimally regularized spline functions seem more appropriate, especially if the second derivatives deviate significantly from zero at the record ends.

The robustness of Utreras' quintic spline estimation algorithm was improved by using Gentleman's (1973) orthonormal triangularization procedure, at the expense of some increase in computational effort. An even better approach would be to use the so-called B-splines which are more stable than the piecewise polynomials used in the present study (Lyche and Schumaker 1973; de Boor 1978; Lee 1982). This could be combined with the sparse matrix procedure described by Eldén (1984), and with Gentleman's procedure. One may note that the latter procedure is sometimes criticized for the numerical overhead incurred by avoiding time-consuming square-root calculations. However, part of this overhead is caused by array addressing operations, which nowadays can occur concurrently with floating-point calculations.

Summarizing, it would appear that optimally regularized, natural quintic spline functions are useful for smoothing and differentiation up to at least the second derivative. Under appropriate boundary conditions, even the third derivative may be reliably estimable, for sufficiently accurate data. Ragozin (1983) recently provided a theoretical analysis of error bounds for derivative estimation in (polynomial) spline smoothing. In essence, differentiation for orders $< m$ is generally feasible, if the function underlying the noisy data is squared integrable over the measurement interval in at least the m -th derivative, and absolutely continuous in this interval for all lower derivatives.

The present paper confines itself to equidistantly sampled data with stationary noise; by consequence, data loss due to, e.g., shadowing effects in movement analysis is not accounted for. Using the algorithm of Hutchinson and de Hoog (1985), optimally regularized spline functions can also accommodate arbitrary sampling distances and nonstationary noise in a numerically efficient manner.

Concluding remarks

In an applied context, the large variety of different solutions claiming to solve a given class of problems can be rather bewildering, especially

if the parent disciplines are wide apart. This is certainly the case in biomechanics, with impacts from anatomy, physiology, mathematics, computer and systems science, engineering, and programming (Hatze 1984). The present problem of estimating smooth derivatives from noisy data makes this quite clear: even the parent disciplines for this particular topic are far apart. 'Digital filtering' [3] tends to be practised by electrical engineers (e.g., Oppenheim and Schaffer 1975; Rabiner and Gold 1975), while regularization and spline fitting are more in the realm of statistics and mathematical physics. Fortunately, the literature shows certain interconnections. The optimally regularizing approach of Anderssen and Bloomfield (1974a, 1974b) is based on the Fast Fourier Transform which is extensively used in electrical engineering and communication, while Craven and Wahba (1979) demonstrated that optimally regularized spline functions for the equidistantly sampled, stationary noise case are closely related to '...the Butterworth filter, well known to electrical engineers'. In Pezzack et al. (1977), the Butterworth filter was found to be most suitable for the authors' kinesiological purposes.

Current studies on internal joint biomechanics suggest the occurrence of high bone-on-bone contact forces, which are typically of the order of 5 times body weight in the lower limbs. These forces are caused by stabilizing co-contractions in agonistic and antagonistic muscles, and by muscular and ligamentous counterforces in order to compensate for off-axis body weight. Since only the net balance of all these forces is estimable through the 'external biomechanics' of observable body motion, it is imaginable that variations in this net balance can have strong influence on the internal forces in the muscles, ligaments, and articulating surfaces. Thus, it seems necessary to ensure that these external forces are accurately determined, in addition to the use of appropriate models and measurements for internal joint biomechanics.

Other applications for accurate derivative estimation in biomechanics include curvature analysis, which requires first and second derivatives of surface co-ordinates. Examples are articulating joint geometry,

[3] The term 'digital filtering' can be viewed as a misnomer since the word refers in particular to implementational aspects. From this point of view, any computerized filtering procedure, including splines, Kalman filters, etc., can be a 'digital filter'. However, the term has acquired established status for linear filters in a frequency/time context. See also Rabiner et al. (1972).

and the shape of the back in the study of scoliosis and kyphosis. For this purpose, one might use optimally regularized, bivariate polynomial splines; see, e.g., Utreras (1979) and Wahba (1979).

The ill-posed nature of the derivative estimation problem has some essential implications for movement synthesis and optimization. Since accelerations are ill-defined by noisy position data, a wide variety of acceleration patterns can yield virtually identical displacements. Thus, the organism might use additional criteria which hardly affect the kinematic purpose of the movement, such as minimizing the occurrence of high force transients (inertial force transients correspond with the third position derivative, or *jerk*). In mechanical engineering, high force transients are often related to tear-and-wear effects, and one might speculate that the biological organism tries to minimize such influences. This is an argument in favour of using even higher-order splines than discussed in this paper (e.g., heptic splines, with $m = 4$). Here, the use of B-splines might prove numerically more advantageous.

The above considerations suggest that equipment for analysis of human movement should have a high resolution in time and space, so as to enable the assessment of any true, high-frequency acceleration transients in the data. In accelerometric studies, such high-frequency components have indeed been found (e.g., Wosk and Voloshin 1981). Finally, it should be mentioned that additional modelling effort is required to counter the various artefacts in human motion analysis, such as caused by skin movement when artificial landmarks are affixed to the body (e.g., Woltring 1983).

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